

**B.Sc. (HONS.) IN CSE FIRST YEAR, SECOND SEMESTER
EXAMINATION, 2021**

DISCRETE MATHEMATICS

[According to the New Syllabus]

Subject Code : 510223

Examination Code : 5612

Time—3 hours

Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any four of the following questions.]

Marks

1. (a) Which of the following sentences are proposition? What are truth values of those that are propositions? 4
 - (i) Savar is the capital of Bangladesh.
 - (ii) Is he ill?
 - (iii) $x + 2 = 5$.
 - (iv) You pass the course.
- (b) Let $P(x)$, $Q(x)$ and $R(x)$ be the statements “ x is a lion”, “ x is fierce” and “ x drinks coffee” respectively. Assuming that the universe of discourse is the set of all creatures. Express each of the following statements using quantifiers, logical connectives and $P(x)$, $Q(x)$ and $R(x)$: 8
 - (i) All lions are fierce.
 - (ii) Some lions do not drink coffee.
 - (ii) Some fierce creatures do not drink coffee.
- (c) Consider the following propositions : 4

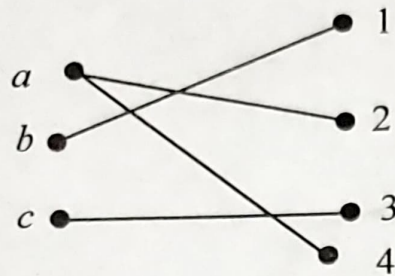
p : You can send an E-mail.
 q : Your inbox is full.
 r : New e-mails are stored in your inbox.

Translate the following sentences into logical expressions :

 - (i) You cannot send an e-mail if and only if your inbox is full.
 - (ii) New e-mails are stored in your inbox unless your inbox is full.
- (d) Show that $(p \wedge q) \longrightarrow (p \vee q)$ is a tautology. 4
2. (a) Define set and power set. What is the power set of $\{a, b, c\}$? 5
- (b) Let $A_i = \{1, 2, 3, \dots, i\}$. Then $\bigcup_{i=1}^n A_i =$ and? $\bigcap_{i=1}^n A_i =$? 5
- (c) Define composition of function. Let f and g be the function from the set of integers to the set of integers defined by the $f(x) = 4x + 2$ and $g(x) = 2x + 1$. What is the composition of f and g ? What is the composition of g and f ? 5

[Please turn over]

- (d) What is function? Is the following figure a function? Why is it or not? 5

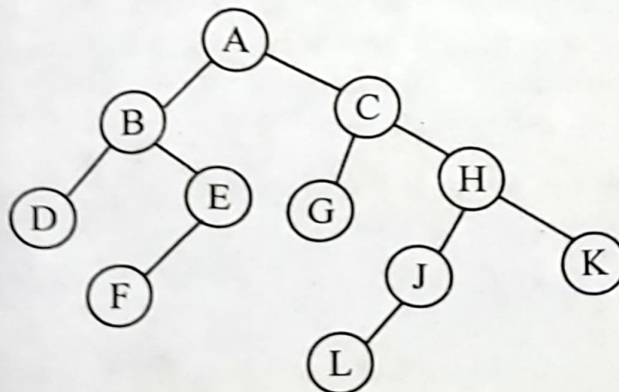


3. (a) What is relation? Write down the properties of relation. 5
 (b) Let $R = \{(1, 1), (2, 1), (3, 2), (4, 3)\}$. Find the powers $R^n, n = 2, 3, 4, \dots$ 5
 (c) Consider the following relation on $\{1, 2, 3, 4\}$:
 $R = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1)\}$. Is the relation reflexive, symmetric and transitive? 5
 (d) Find the matrix representation of the relations SoR where the 5

matrices representing R and S are $M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and

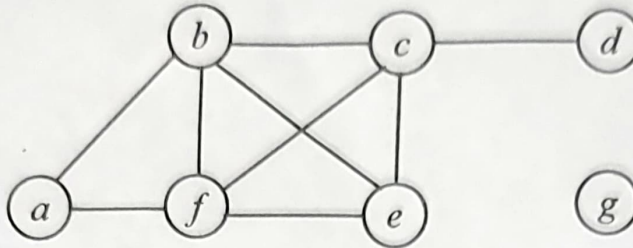
$$M_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

4. (a) Draw the Hasse diagram representing the partial ordering $\{(a, b) \mid a \text{ divides } b\}$ on $\{1, 2, 3, 4, 6, 8, 12\}$. 5
 (b) Draw a binary tree from the traversal : 5
 Inorder : E A C K F H D B G
 Preorder : F A E K C D H G B
 (c) Write down the preorder, inorder and postorder traversing for the 5
 following binary tree :



- (d) Evaluate the prefix expression $+ - * 235 / \uparrow 234$?

5. (a) What is graph? What is the degree of vertices in the graph G displayed in following figure : 5



- (b) Draw the precedence graph for the following expression : 5

S1 : $x := 0$

S2 : $x := x + 1$

S3 : $y := 2$

S4 : $z := y$

S5 : $x := x + 2$

S6 : $y := x + z$

S7 : $z := 4$

- (c) Define chromatic number. What is the chromatic number of $K_{3,4}$ and K_5 ? 5

- (d) Draw a graph with the adjacency matrix : 5

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

6. (a) What is mathematical induction? Using mathematical induction prove that the sum of the first n odd positive integers is n^2 . 5

- (b) Define the Sum Rule and the Product Rule. How many different bit strings are there of length eight? 5

- (c) Describe the universality of NAND and NOR gate. 5

- (d) Simply the Boolean function $F = \overline{ABC} + \overline{BCD} + \overline{ABCD} + \overline{ABC}$ using K-map. 5